Common questions in Data Preprocessing

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Question 1

SHOULD YOU SPLIT THE DATASET OR SCALE FEATURES FIRST?

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Question 1:
Should You Split the Dataset or Scale Features First?

- Answer:
  - apply feature scaling after splitting the dataset into the Train set and Test set
- Explanation: Why Not Scale Before Splitting?
  - If you scale before splitting:
    - ❌ You use information from the test set
    - ❌ This causes data leakage
    - ❌ Your model performance becomes fake (over-optimistic)

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Question 1:
Should You Split the Dataset or Scale Features First?

- Simple Analogy
  - Think of it like an exam:
    - Training set = studying
    - Test set = exam
- If you look at the exam before studying → ❌ cheating

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Question 2

SHOULD DUMMY VARIABLES BE SCALED OR STANDARDIZED?

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Question 2:
Should dummy variables be scaled or standardized?

- **Question:**
  - do we have to apply feature scaling, Standardization to the dummy variables in the matrix of features?
- **Answer:**
  - No, dummy variables are already binary and on the same scale, so scaling them is unnecessary and may reduce interpretability.
- **Key Rule:**

Feature Type	Apply Feature Scaling?
Numerical (age, salary)	✅ Yes
Dummy variables (0/1)	❌ No

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Question 2:
Should dummy variables be scaled or standardized?

- The Exception:
  - there is one specific case where people sometimes do it anyway:
    - Distance based algorithms (like KNN or SVM).In those specific cases, if you don't scale the dummy variables, the "distance" between 0 and 1 might be treated differently than the distance between other scaled continuous features (which often end up between -1 and 1 or 0 and 1). However, for most models (like Linear Regression, Trees, or Neural Networks), leaving them as 0 and 1 is the standard and preferred practice to keep the model interpretable.

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Question 3

SHOULD YOU SCALE Y (TARGET VARIABLE) OR ONLY X (FEATURES)?

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Question 3:
Should you scale y (target variable) or only X (features)?

- Answer:
  - We scale X (features) only, not y (target) in most cases.

- The Exception:
  - If your target values y are extremely large (e.g., predicting multi million dollar$ house prices) or have a very wide range, scaling y can help the model converge faster during training. However, if you scale y, you must remember to inverse transform the predictions at the end to get the real-world values back.

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Summary Table

QUESTION 2 & QUESTION 3 SUMMARY

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Summary Table:
Question 2 & Question 3 Summary

- Summary Table:

Component	Scale?	Reason
Features (X)	Yes	Prevents features with large units from dominating the model.
Dummy Variables	No	They are already 0 and 1 (on a small, fixed scale).
Target (y)	No	Keeps the output readable and interpretable.

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Small Tips

FEATURE SCALING TIPS

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Small Tips:
Feature Scaling tips

- Standardization → used in most ML models (like SVM, Logistic Regression)
- Normalization → useful when data has fixed range or for neural networks
- My recommendation always use Standardization

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Note:

TRAIN-TEST SPLIT

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Note:
Train-Test Split

- When you split the data, you don’t just get two sets
  - you actually get four:
    - X train, X test
    - y train, and y test

Follow for More

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- Simple explanations + Practical examples

Follow for more

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